

# **Selection assay to identify desiccation tolerant bacteria.**

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## **Abstract**

We describe a method to evaluate bacterial tolerance to air desiccation at 30 °C and 50% relative humidity.

## **Growth medium**

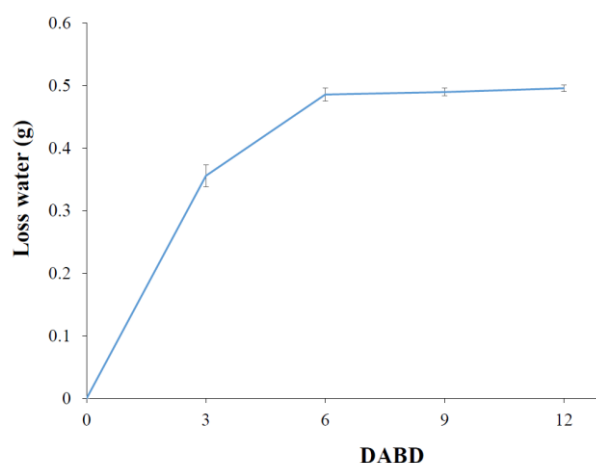
SYP broth is used for the cultivation of different bacteria until stationary phase. The composition of this medium is sucrose 10g, yeast extract 1g, K<sub>2</sub>HPO<sub>4</sub> 1g and KH<sub>2</sub>PO<sub>4</sub> 3g, a final pH of 6.2[1,2].

## **Description of the method**

Bacteria are grown until stationary phase in SYP liquid medium, because at this growth stage cells are able to resist stresses better than cells in exponential phase [3]. Afterward, bacterial cells are harvested by centrifugation, washed twice with sterile water, and resuspended in the initial volume (45 ml) of sterile water. Before desiccation, the number of cultivable cells is quantified by quintuplicate using the massive stamping drop plate (MSDP) in SYP medium plates [4]. For each strain, thirty sterile polypropylene tubes are filled with 500 µL of the bacterial suspension and covered with a sterile cotton cap. The tubes with the bacterial suspension are placed in a desiccation chamber at 30 °C and 55 % relative humidity (RH). Under those conditions water of each tube is completely loosed

after six days and this is corroborated by the constant weight of the tubes (an example is showed in Figure 1).

We recommend the counting of cultivable cells, after rehydration of desiccated cells, every three days for a period of twelve days using the MSDP method. For the samples of 3 days after the beginning of desiccation (DABD) the tubes are filled with water to the initial 500  $\mu$ L. For completely desiccated samples (6, 9, and 12 DABD), 500  $\mu$ L of water must be added to each tube and after 40 minutes with occasional shaking, cell re-hydration must be completed. The bacterial survival ratio (BSR) is reported as the ratio of the log number of bacterial cells present in the suspension at any time after the beginning of desiccation (AbD) plus one to the log number of viable cells before desiccation (BD), multiplied by 100, i.e.,  $BSR = (\log_{AbD+1} / \log_{BD}) \times 100$  [3]. The BSR is a parameter that allowed us to quantify the desiccation tolerance of bacteria.



**Figure 1.** Water loss in *Sphingomonas* sp. OF178 samples subjected to desiccation at 30 °C and 50% HR. DABD means Days After the Beginning of Desiccation.

## References

1. Caballero-Mellado J, Martinez-Romero E. Limited genetic diversity in the endophytic sugarcane bacterium *Acetobacter diazotrophicus*. Appl Environ Microbiol. 1994;60: 1532–1537.
2. Tapia-Hernández A, Bustillos-Cristales MR, Jiménez-Salgado T, Caballero-Mellado J, Fuentes-Ramírez LE. Natural endophytic occurrence of *Acetobacter diazotrophicus* in pineapple plants. Microb Ecol. 2000;39: 49–55. doi:10.1007/s002489900190
3. Muñoz-Rojas J, Bernal P, Duque E, Godoy P, Segura A, Ramos JL. Involvement of cyclopropane fatty acids in the response of *Pseudomonas putida* KT2440 to freeze-drying. Appl Environ Microbiol. 2006;72: 472–477. doi:10.1128/AEM.72.1.472-

477.2006

4. Morales-García YE, Corral-Lugo A, Pazos-Rojas LA, Martínez-Contreras RD, Muñoz-Rojas J, Ramírez-Valverde A. Cuantificación de bacterias cultivables mediante el método de “Goteo en Placa por Sellado (o estampado) Masivo.” Rev Colomb Biotecnol. 2012;14: 147–156.